

CS1-TYRE ROAD WEAR PARTICLES (TRWP)

Goal: Design optimal passenger car tyre tread compound that **reduces TRWP emissions**, maintains/improves key tyre performance metrics such as **wear resistance**, **rolling resistance (fuel economy)**, and **wet grip**, and finally **minimises the environmental/health impacts of materials used**

- Design the optimum passenger car tyre tread compounds that improves tyre wear properties.
 - Whats the problem?
 - What kind of abrasion process happens with the tyres?
 - Focus on which polymer or polymer blend and which fillers or other additives are required.

Whats the problem?

TRWPs are **emitted due to the friction** between **vehicle tyres and the road surface** during driving, braking, and cornering. They are a **mixture of tyre tread material and pavement material**, in **equal parts**. They are typically **cigar-shaped**, generally below 1mm in length and typically between 70-100um. The average density is about **1.8 Mg/m³**

These particles are one of the most significant contributors to **non-exhaust emissions** of **microplastics** in the environment. They enter air, water, and soil, potentially harming both ecosystems and human health.

The toxicological profile of Tyre Road Wear Particles (TRWPs) reveals they trigger a concentration-dependent **proinflammatory response**, as shown by increased **TNF-α production** in human cells. Additionally, TRWP leachates have been found to be **acutely toxic to aquatic life**, particularly adult coho salmon, largely due to compounds like **6PPD-quinone**, a transformation product of a common tyre antioxidant. These findings highlight growing environmental and health concerns associated with TRWP exposure.

TARGET: Reduce the impact of materials that are harmful to health or the environment → Reduce such additives as much as possible

What kind of abrasion process happens with the tyres?

TYRE WEAR IS PROPORTIONAL TO FRICTIONAL ENERGY

Wear $\propto$ Friction Energy = Friction Force $\times$ Sliding Distance ← depends on tyre design and frictional force.

And friction force is related to:

Friction force = $\text{MASS}_{\text{vehicle}} \times \text{Acceleration of vehicle (longitudinal + lateral)}$

Heavier vehicles and aggressive driving (rapid acceleration, hard braking, sharp cornering) **increase frictional energy**, leading to **greater tyre wear** and **more Tyre Road Wear Particles (TRWPs)**. *This effect is also influenced by tyre design and material choice, which control how much friction translates into actual wear*

TYPES OF WEAR: SMEAR and ABRASION

SMEAR WEAR-low severity wear	ABRASIVE WEAR-high severity wear
Seen in: Conventional vehicles, normal road use. Appearance: smaller micro crack formations occur in much higher frequency, -no abrasion pattern Mechanism: Tribo chemical degradation: "An oxidised sticky layer protects the bulk rubber surface from excessive crack initiation and propagation." Effect: Generates larger, softer particles that may stay on or near the road surface. Appearance: smaller micro crack formations occur in much higher frequency, generates 1/10th vol abrasion-no abrasion pattern Desirable for TRWP reduction because: • Less aggressive material removal. • Fewer sharp or airborne particles. Smear wear happens when the rubber compound can deform plastically or viscoelastically, absorbing frictional energy without cracking.	Seen in: High-performance vehicles, aggressive driving. Mechanism: Driven by thermo-mechanical degradation: "Accelerated particle detachment and abrasive (and fatigue) wear." Appearance: Crack formations develop on the macro scale at evenly spaced intervals. Distinct rough patches, tearing or pitting on the surface. - small abrasion pattern. (road surfaces act like cutting tools-generate more debris) Effect: Produces smaller, sharper particles— higher risk of airborne PM2.5 or PM10. Leads to more TRWP emissions and faster tread loss.

Design the optimum passenger car tyre tread compounds that improves tyre wear properties. Focus on which polymer or polymer blend and which fillers or other additives are required.

To **improve tyre wear performance** and reduce **Tyre Road Wear Particles (TRWP)**, the compound must encourage **smear wear** (not abrasive wear). This requires a **tough, elastic, and thermally stable** formulation.

Selection of fillers affected by:

SIZE:

- **Smaller particles with higher specific surface area** (e.g. N220, N226) reduce particle detachment.
- Improve filler–rubber interaction → enhance wear resistance.

SHAPE: branched shapes better-less wear

SURFACE ENERGY: **functionalized fillers** to promote dispersion and bonding.

Curing:

Final tyre tread composition:

Polymer base:

- **Natural Rubber (NR) – 70 phr**
Provides **high elasticity** and **low energy loss**(as **rolling resistance is low**), which encourages **smear wear**, reducing TRWP formation (highlight:low rolling resistance.)
- **End-Modified Solution SBR (SSBR) – 30 phr**
Enhances **wet traction**, improves **thermal stability**, and promotes strong **filler interaction** with silica. Also contributes to toughness and wear resistance.(provides good grip)

The **bi-continuous phase and nano-structured phase structure** of NR and filled SSBR ensures **high toughness** with **balanced viscoelastic properties**.(wet grip +rolling resistance improved)

Fillers:

- **Silica (50 phr, high surface area)**
Reduces rolling resistance and improves wet grip. Works with functionalised polymers.
- **Carbon Black (5 phr, high structure)**
Included in small quantity for enhanced UV resistance and additional reinforcement-contributes to wear resistance.

Coupling Agent (5 phr, e.g. TESPT)

Improves bonding between silica and SSBR, enhancing **compound durability** and reducing **energy loss**

Processing aids and additives:

- **TDAE Oil**
A non-toxic plasticizer that softens the compound for easier processing without environmental harm.
- **Alternative 6PPD**
Replaces traditional antioxidant (6PPD), reducing **aquatic toxicity** while maintaining oxidative stability.
- **Alternative ZnO**
An eco-friendly replacement for zinc oxide used in vulcanisation activation, reducing metal contamination.

Encourages **smear wear** → reduces TRWP generation.Maintains **wet grip and rolling resistance** performance.Uses **eco-friendlier additives for improved sustainability**.Supports **high toughness and elasticity** via a bi-continuous NR–SSBR structure.

- Describe how you would mix the rubber compound and then prepare it for moulding into a car tyre.
 - How do we design tyre compounds
 - How do we understand the process?

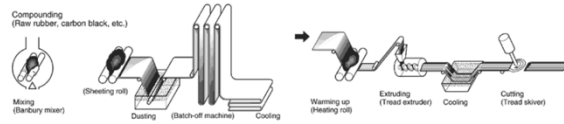
Selection of Polymers

* NR: natural rubber, IR: Isoprene Rubber, SBR: Styrene Butadiene Rubber, BR: Butadiene Rubber

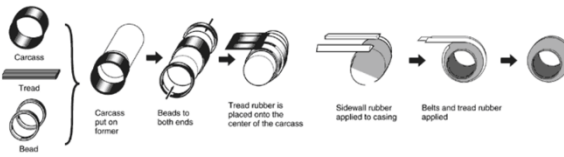
	NR	IR	SBR	BR
Wear Resistance	⊙	○	⊙	☆
Wet Traction	○	○	⊙	○
Energy Loss (RR)	⊙	☆	○	×
Tear Resistance	☆	⊙	×	×
Air permeability	×	×	×	×
Heat Degradation Resistance	×	×	⊙	○

☆Super, ⊙ Superior, ○ Fair, × Inferior

Mixing and extruding process



Assembling process



Vulcanizing process



1. Mixing (Banbury mixer)

Raw rubber is combined with fillers like carbon black, silica, oils, and additives in a high-shear internal mixer. This produces a uniform rubber compound ready for shaping.

Sheeting and Cooling (Batch-off machine)

The mixed rubber is rolled into flat sheets and cooled down. These sheets are dusted to prevent sticking before further processing.

2. Extruding (Tread extruder)

The cooled rubber is forced through a die to create specific shapes like the tread profile. It is then cooled and cut into set lengths.

3. Assembling process

Carcass and bead assembly

A drum is used to build the tyre's core (carcass), onto which **beads** are placed at both ends for rim locking. This forms the structural backbone of the tyre.

Adding tread and sidewalls

Pre-shaped rubber for the tread and sidewalls is applied over the carcass. Steel belts may be added on top for reinforcement and strength.

Final green tyre assembly

The unvulcanized tyre, called a "green tyre," is now fully built but still soft and not ready for use. It's transferred for curing.

4. Vulcanizing process

Moulding and heating

The green tyre is placed into a heated mould that presses and shapes it under high pressure. The heat activates sulfur, cross-linking the rubber molecules (vulcanization).

Finishing and inspection

After vulcanization, the tyre is cooled, removed from the mould, and trimmed. It undergoes quality checks (e.g., balancing, x-ray, visual) before being approved for use.

TYRE COMPOUNDS: The compound structure follows:

- **Step 1:** NR / Filled f-SSBR blend
- **Step 2:** Formation of bi-continuous morphology
- **Step 3:** Uniform curing for energy-efficient, durable performance

Low-temperature vulcanisation forms a **uniform crosslinked network**, further lowering $\tan \delta$ and increasing toughness.

Explain how your tyre would perform in the other tyre performance measures such as enhanced fuel economy or emergency braking tests.

High loading filler gives better wear and saturation of energy loss/ WET grip

Less loading filler gives better energy loss and saturation of wear

Hysteresis is the **energy lost as heat** when a tyre is repeatedly deformed and recovered during rolling. The **type and composition** of rubber used in the tyre affects its **viscoelastic properties**, and therefore the amount of **hysteresis** generated.

- **High hysteresis** → Better **wet grip and traction** (great for **braking**), but more energy loss.
- **Low hysteresis** → Lower **rolling resistance** (better **fuel efficiency**), but less grip.
- **Wet grip** demands high energy loss (high $\tan \delta$) at **high frequency** (cold temps, sudden braking).
- **Rolling resistance** demands low energy loss (low $\tan \delta$) at **low frequency** (warm temps, steady cruising).

Time temperature superposition

- **Low temperature $\approx$ High frequency** → simulates braking and wet traction.
- **High temperature $\approx$ Low frequency** → simulates long, slow rolling.

Can **predict behaviour** during driving different driving conditions-use it to optimise tyre formulation, using compounds and nanostructure-according to magic triangle.

NANOSTRUCTURE BLEND

Trade-off problem between good wear resistance and enhanced wet grip/toughness lends to a bi-continuous polymer blend solution:

→ Filled SBR; for high grip (high toughness and higher hysteresis)

High loading filler gives better wear resistance and saturation of energy loss (hysteresis) / wet grip

→ Unfilled NR; some good aspects of abrasion resistance with very low rolling resistance (lower energy loss and higher toughness)

This mitigates the compromise between wear resistance and rolling resistance and incorporates both.

CHAIN END MODIFICATION

In general, fillers are needed to enhance performance. However, in a bi-continuous phase such as this one, SBR needs fillers to enhance wet traction whereas NR should avoid fillers to preserve lower hysteresis.

To mitigate this, SBR is functionalised so that it contains end groups which are more attractive and therefore the fillers stick to the SBR phase, and not the NR phase.

Enhanced fuel economy (rolling resistance)

The tyre uses **unfilled Natural Rubber (NR)** and **high-surface-area silica** in a **bi-continuous blend** with **filled, functionalised SSBR**. This structure, along with **low-temperature vulcanisation**, reduces energy loss (low $\tan \delta$) at **low frequencies**, reducing rolling resistance. This improves **fuel efficiency** by reducing fuel consumption.

- **Low hysteresis** → **Less energy lost** → **Better fuel economy**
- **Nanostructure control + good filler dispersion** = optimised performance

Braking-wet grip

Under sudden braking (wet or cold conditions), tyres experience **high-frequency deformation**. This tyre formulation includes **end-modified SSBR**, which chemically bonds to silica via **coupling agents** (e.g., TESPT), forming a strong filler-polymer network. The **high filler loading** in SSBR boosts energy loss at high frequency ($\tan \delta$)=increases friction at higher frequencies=shorter stopping distance.

- **High hysteresis at high frequency** = **better wet grip**
- **Chain-end functionalisation** of SSBR ensures better dispersion and grip
- **Bi-continuous structure** optimises wet grip and fuel economy.

Other notes:

Don't want any low friction coating on tyre, any surface coating apply to surface will wear off very fast-need a surface that's same as rest of tread block.

